# Training-Free Off-Screen Player Imputation for Broadcast-Based Spatial Football Analytics

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**Abstract**

Spatial football metrics such as pitch control assume access to the positions of all 22 players, yet the most widely available source of positional data — the broadcast main camera — shows only 10–16 of them at any moment. We quantify the resulting distortion with an open, reproducible benchmark: a simulated broadcast viewport applied to open full-pitch tracking data (Metrica Sports; three matches, one held out from method development). Ignoring off-screen players — the visible-only baseline implied whenever a video-based game-state-reconstruction (GSR) pipeline adds no imputation layer — inflates hidden-zone pitch-control error to 25.1–26.9 percentage points and produces a mean absolute control-share error of 11.1–13.4 points across the three matches. We then evaluate a ladder of *training-free, online* imputation baselines that use only observations from the match being analysed. The best overall on these decision-relevant metrics, *role-anchored centroid voting* — each visible player votes for the full-team centroid by subtracting its running role offset, attenuating the viewport-induced subset bias — roughly halves hidden-zone error (to 12.2–13.8 points) and cuts the control-share error to 28–48% of the ignore policy at every viewport width from 36 m to 60 m in all three matches (4.5–4.7 points at $W = 44$ m). In the short-occlusion regime ($\leq$9.6 s) covered by the closest learned prior work, our training-free method reaches binwise median position errors of 3.3–8.9 m; 50–57% of hidden-player observations under the simulated viewport, however, lie beyond that regime, outside the 9.6 s sequence protocol of that prior work. We integrate the method end-to-end into an open broadcast-video GSR pipeline and show that imputation changes a downstream possession-quality score (Space-Creation Index) by 15.6 and 17.2 points on two real World Cup broadcast windows, flipping the verdict class — under prespecified operational thresholds — in one of them.

## 1 Introduction

Video-based game state reconstruction (GSR) turns ordinary broadcast footage into player coordinates on a common pitch template, promising tracking-level analytics without stadium hardware [12]. But the broadcast main camera pans and zooms with the ball: in our World Cup broadcast clips only 10–16 of the 22 players are visible in a typical frame, consistent with the 9–12 reported for broadcast video by Ochin et al. [9] and the 12.8±3.7 of Omidshafiei et al. [10]. Team-level spatial metrics — pitch control [13, 14], space value, block compactness — are then computed over a biased subset of players: the team whose defenders sit off-screen silently cedes control of the hidden half of the pitch, not because of anything that happened on the pitch, but because of where the camera pointed.

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This distortion is routinely acknowledged and rarely measured. The GSR evaluation protocol itself (GS-HOTA [12]) scores only the players that are visible; what happens to downstream *team-level* metrics when the invisible remainder is ignored has, to our knowledge, not been quantified on open data. The one closely related study, DeepMind's Graph Imputer [10], demonstrated the problem and a learned solution, but on 105 proprietary Premier League tracking matches, with a bidirectional model that consumes *future* observations inside 9.6-second windows — neither reproducible nor applicable in a strictly online setting.

This paper makes three contributions:

1. **A reproducible distortion benchmark.** We simulate a broadcast viewport (ball-tracking pan with lag, width $W$) over open full-pitch tracking data and score any imputation policy by hidden-zone pitch-control MAE, team control-share error, and hidden-player position error — metrics defined by the downstream analytics rather than by trajectory fidelity alone (Section 3).

2. **A ladder of training-free online baselines**, culminating in *role-anchored centroid voting* (B4), which halves the hidden-zone control error and cuts control-share error to 28–48% of the ignore-policy baseline, without offline training, future observations, or GPU (Sections 4–5).

3. **End-to-end integration** into an open video→GSR→metrics pipeline, with a case study on real World Cup broadcasts where imputation moves a possession-quality score by a decision-relevant magnitude, changing the verdict class in one of two windows (Section 6).

Our position is deliberately modest: learned imputers remain preferable when training data and future context are available. What we establish is (i) how large the distortion actually is under a simulated viewport, (ii) a training-free floor that learned methods should beat, and (iii) an explicit scoring of the long-occlusion regime (>9.6 s) that lies outside the sequence protocol of the closest prior work — and turns out to hold roughly half or more (50–57%) of hidden-player observations in our benchmark.

## 2 Related Work

**Learned trajectory imputation.** The Graph Imputer of Omidshafiei et al. [10] is the closest prior work: a graph-network variational model trained on 105 proprietary EPL tracking matches, evaluated with a simulated camera view-cone, and applied to pitch control under partial observability — the same problem framing as ours. Two protocol differences preclude direct numeric comparison, and we analyse them explicitly in Section 5.1: their model is bidirectional (it consumes future observations within 9.6 s windows), and trajectories are segmented into 9.6 s sequences — bounding the occlusion span the model must bridge — whereas our setting is strictly online with unbounded occlusions. MIDAS [3] is a recent learned set-transformer imputer reporting strong results on multi-agent trajectory imputation, including camera-mask experiments; TranSPORTmer [2] similarly unifies forecasting, imputation, and agent-status inference for multi-agent sports in a single transformer under an input mask. Everett et al. [4] impute full-team player locations from sparse event observations using recurrent and graph-based components, enabling downstream analyses such as pitch dominance; their observation setting is event-driven rather than camera-mask-driven and requires offline training, whereas our benchmark targets causal broadcast-view occlusion without training data. Penn et al. [11] reconstruct continuous trajectories from discrete broadcast-derived observations with a parsimonious autoregressive (ARMAX) model that uses the ball position as an exogenous input — an adjacent observation model where sparsity is temporal rather than spatial.

**Ghosting.** The ghosting line of work [7, 15] generates counterfactual league-average positions — where a player *should* have been — for coaching evaluation. The goal differs from ours: ghosting evaluates decisions against a baseline policy, whereas imputation recovers actual unobserved positions for measurement.

**Game state reconstruction.** SoccerNet-GSR [12] defines the task and the GS-HOTA metric for recovering *visible* players from broadcast video; camera calibration [5, 6] and multi-object tracking [1] supply the geometry and identity backbone. FOOTPASS [9] reports 9–12 visible players per broadcast frame and notes imputation as a pipeline necessity, corroborating the universality of the problem. Our benchmark scores the invisible remainder that GS-HOTA leaves undefined.

# 3 Benchmark

**Data.** Metrica Sports open tracking [8]: three matches, all 22 on-pitch players plus ball at 25 fps (games 1–2 ship as full-match CSVs; game 3, in the EPTS-FIFA format, covers one half). We use the first 45 minutes of each and evaluate at 5 fps; individual player samples with missing coordinates are dropped (frames are kept). Games 1–2 were used for method and hyperparameter development; game 3 — and the ablation variants B3E/B3V/B5 — were specified and frozen before game 3 was first evaluated, so game 3 serves as a held-out check of the method ladder (the interpretive text was, of course, written after seeing all results). Using full-pitch tracking as ground truth lets us compute the *true* spatial metrics and score any partial-observation policy against them.

**Viewport simulator.** A virtual main camera pans horizontally following an exponentially smoothed ball position ($\alpha = 0.06$ per frame at 25 fps), mimicking broadcast pan lag; the visible region is a width-$W$ window spanning the full pitch height. At $W = 44$ m the viewport shows 14.6–15.0 players on average (games 1–3), within the range of our real broadcast clips (10–16 visible) and close to the 12.8±3.7 reported by Omidshafiei et al. [10]. Sensitivity is reported for $W \in \{36, 44, 52, 60\}$ m.

**Metrics.** We score three quantities, chosen to reflect the downstream analytics rather than trajectory fidelity alone:

1. **Hidden-player position error** (m): distance between imputed and true position, over all (frame, player) pairs where the player is outside the viewport and has been observed at least once earlier in the half (never-observed players cannot be scored for position; they remain absent from the control maps of every partial-observation condition — the ground-truth map always contains them — so the map metrics do include the cold-start penalty). Occlusion gap denotes elapsed time since the player's last observation at each hidden frame; occlusion-share denominators use this same population.

2. **Pitch-control map MAE**: pitch control is computed on a 3 m grid with an arrival-time model and sigmoid contest function [14]; we report mean absolute error against the full-observation control map, over the full pitch and over the *hidden zone* (grid cells outside the viewport) separately. Control is computed with zero velocities in all conditions, isolating the effect of imputation from that of velocity estimation; the reported magnitudes therefore apply to this position-only control variant (velocity-aware control is discussed in Section 7).

3. **Team control-share error** (percentage points): mean absolute per-frame deviation of the single headline number "team A controls $x$% of the pitch".

We designate (1) and (3) as the *decision-relevant* metrics — they determine where players are placed and what share is reported — and select among methods on them; the map MAE (2) is reported for completeness. The hidden-zone MAE and the share error carry 95% block-bootstrap confidence intervals over one-minute blocks (frames within a block are strongly autocorrelated, so frame-level resampling would be anticonservative); these within-match intervals quantify temporal sampling uncertainty, not match-to-match variation.

# 4 The Imputation Ladder

All methods below are *causal* (no future observations), use only observations from the current match (no offline training), and run faster than real time on a single CPU core (closed-form arithmetic, no learned components). Let $V_t$ denote the set of visible players at time $t$, $p_i(t)$ the position of player $i$, and $\mathrm{off}_i(t)$ a running estimate of player $i$'s *role offset* — its displacement from the team centroid. All quantities are maintained separately per team (including the three-voter threshold below); we drop the team index for readability.

**B0 — ignore.** Hidden players are absent from the metric — the visible-only baseline implied whenever no imputation layer is added; GSR evaluation itself covers visible players only [12].

**B1 — last-seen with decay.** $\hat{p}_j = w\, p_j^{\text{last}} + (1-w)\,\bar{p}_V$ with $w = e^{-\Delta t/\tau}$, $\tau = 8\,\text{s}$: hold the last observed position, decaying toward the visible-team mean.

**B2 — formation anchor.** While player $i$ is visible, store $\mathrm{off}_i = p_i - \bar{p}_V$, its offset from the *visible-team* centroid (a visible-subset offset — the viewport bias it inherits is what B4 removes). When hidden, place the player at the current visible centroid plus the stored offset.

**B5 — fixed formation template.** As B2, but the offset is the cumulative mean over all of player $i$'s visible frames so far, rather than the latest snapshot — the closest online analogue of a static role template.

**B3 — B2 + EMA offsets + velocity extrapolation.** As B2, but with exponentially averaged offsets — stored, like B4's, relative to the voted centroid of Eq. (1) when available, yet *applied* at the plain visible centroid — and constant-velocity extrapolation for recently hidden players, blended into the anchor with weight $e^{-\Delta t/1.5\,\text{s}}$. Its two ingredients are isolated as B3E (EMA only) and B3V (velocity only) in the ablation (Section 5).

**B4 — role-anchored centroid voting.** The visible-team centroid $\bar{p}_V$ in B2 is itself biased by the viewport: when the camera points left, the visible subset sits left of the true team centroid, and every stored offset inherits that bias. B4 attenuates it by *voting*: each visible player proposes the full-team centroid as its position minus its role offset,

$$\hat{c}(t) \;=\; \frac{1}{|V_t|}\sum_{i\in V_t}\big(p_i(t) - \mathrm{off}_i(t^-)\big), \qquad \mathrm{off}_i(t) \;\leftarrow\; \mathrm{EMA}\big[\,p_i(t) - \hat{c}(t)\,\big] \quad (i \in V_t), \tag{1}$$

where $\mathrm{off}_i(t^-)$ denotes the offset value *before* the update at $t$: each step votes with the previous offsets, then updates them against the voted centroid (EMA weight 0.1 per 5 fps evaluation step). This is a self-consistent bootstrap — offsets are stored relative to the voted centroid,

Table 1: Imputation ladder at $W = 44$ m (first halves of the three Metrica matches; game 3 held out from method development). Lower is better throughout; bold marks the best method per game and metric. B0 has no position error since it places no players.

| | hidden ctrl MAE (pp) g1 / g2 / g3 | full-pitch MAE (pp) g1 / g2 / g3 | share err. (pp) g1 / g2 / g3 | median pos. err. (m) g1 / g2 / g3 |
|---|---|---|---|---|
| B0 ignore | 26.9 / 25.6 / 25.1 | 18.6 / 17.2 / 16.7 | 13.4 / 12.5 / 11.1 | — |
| B1 last-seen | 22.1 / 20.0 / 19.5 | 16.2 / 14.2 / 13.5 | 10.6 / 9.5 / 8.2 | 19.6 / 17.9 / 18.4 |
| B2 anchor | 15.7 / 14.6 / 14.3 | 12.2 / 10.8 / 10.3 | 6.2 / 5.4 / **4.4** | 13.6 / 12.8 / 12.5 |
| B5 fixed template | 23.0 / 18.8 / 19.1 | 17.6 / 14.2 / 14.2 | 10.3 / 7.7 / 6.9 | 22.7 / 20.7 / 21.9 |
| B3E ablation: EMA only | 13.2 / 12.2 / 13.6 | 10.8 / 9.8 / 10.5 | 5.8 / 4.8 / 5.0 | 15.7 / 15.2 / 16.3 |
| B3V ablation: velocity only | 15.5 / 14.4 / 14.0 | 12.0 / 10.6 / 10.0 | 6.2 / 5.5 / **4.4** | 13.2 / 12.2 / 11.9 |
| B3 EMA + velocity | **12.8** / **11.8** / **13.2** | **10.4** / 9.3 / 10.0 | 5.7 / 4.6 / 4.7 | 14.6 / 14.0 / 15.1 |
| B4 centroid vote | 13.3 / 12.2 / 13.8 | 10.5 / **9.2** / **9.9** | **4.7** / **4.5** / 4.7 | **11.6** / **10.0** / **9.7** |

and the voted centroid is computed from those same offsets. The recursion is seeded from the visible mean until three offset-bearing players are available; thereafter, newly appearing players are initialised relative to the voted centroid. Hidden player $j$ is imputed at $\hat{c}(t) + \mathrm{off}_j$. Because each player's individual viewport bias is subtracted before averaging, the systematic subset bias is attenuated rather than accumulated (exact cancellation would require stable offsets and conditionally representative voters). When fewer than three voters are available, B4 falls back to the formation anchor (B2), and B2 in turn to last-seen (B1) for players without a stored offset; such fallback frames are included in the reported numbers. At $W = 44$ m the voting path produces 70–86% of B4's hidden-player estimates (B2 fallback 14–30%, B1 0%), so the reported performance is a mix in those proportions.

## 5 Results

**Ladder.** Table 1 reports the ladder at $W = 44$ m. Ignoring off-screen players (B0) leaves a hidden-zone control MAE of 25.1–26.9 pp and leaves a mean absolute control-share error of 11.1–13.4 pp. The naive last-seen policy removes only about a fifth to a quarter of it. The formation anchor (B2) is the single largest step — most of a hidden player's position is explained by "the team moved, the player kept its role". Centroid voting (B4) then attenuates the viewport subset bias that B2 inherits. It is the best method on position error in all three matches (9.7–11.6 m median vs. 12.5–13.6 m for B2) and on share error in games 1–2 (4.5–4.7 pp); in the held-out game 3, B2 and B3V edge it on share error by 0.3 pp (4.4 vs. 4.7) — a paired one-minute block-bootstrap 95% interval for the game-3 B4−B2 share-error difference is $[-0.4, +1.3]$ pp and includes zero, so this gap is not resolved at this sample size. Full-pitch MAE (Table 1) improves in step. Marginal block-bootstrap 95% intervals for B0 vs. B4 do not overlap in any match on hidden-zone MAE or share error (e.g., game 1 hidden-zone MAE: B0 $[24.5, 29.4]$ vs. B4 $[11.2, 15.6]$; share error $[11.4, 15.7]$ vs. $[3.6, 6.1]$). Figure 2 visualises the effect on a single frame: the hidden zone misassigned by B0 is largely restored by B4.

**Ablation: what actually helps.** The B3E/B3V decomposition settles what drives B3 — and corrects a reading we ourselves held from the confounded B2-vs.-B3 comparison. Velocity blending alone (B3V) *slightly improves* position error over B2 in all three matches (13.6→13.2, 12.8→12.2, 12.5→11.9 m median): short-gap extrapolation is mildly helpful, not harmful. EMA offsets applied at the plain visible centroid (B3E) give the second-lowest hidden-zone map MAE in all three matches

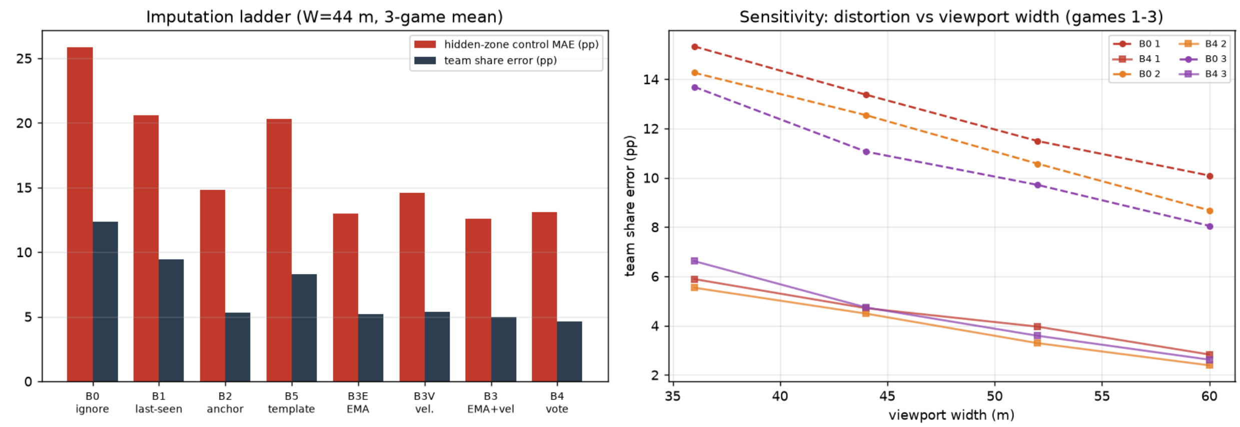


Figure 1: Left: imputation ladder at $W = 44$ m — three-game mean of hidden-zone control MAE and control-share error, all eight methods. Right: viewport-width sensitivity of the share error — B4 (solid) vs. B0 (dashed) across $W = 36$–60 m, all three games. B4's absolute share-error gain is largest in the narrowest (hardest) viewports.

Table 2: Viewport-width sensitivity (game 1 / game 2 / game 3): hidden-zone control MAE and control-share error, B0 vs. B4.

| | | B0 ignore | | B4 centroid vote | |
|---|---|---|---|---|---|
| $W$ (m) | visible (of 22) | MAE (pp) | err. (pp) | MAE (pp) | err. (pp) |
| 36 | 13.0 / 13.1 / 13.1 | 28.9 / 27.6 / 28.3 | 15.3 / 14.3 / 13.7 | 15.6 / 14.1 / 16.8 | 5.9 / 5.5 / 6.6 |
| 44 | 14.6 / 14.9 / 15.0 | 26.9 / 25.6 / 25.1 | 13.4 / 12.5 / 11.1 | 13.3 / 12.2 / 13.8 | 4.7 / 4.5 / 4.7 |
| 52 | 16.1 / 16.3 / 16.5 | 24.4 / 23.0 / 23.3 | 11.5 / 10.6 / 9.7 | 11.6 / 9.8 / 12.1 | 4.0 / 3.3 / 3.6 |
| 60 | 17.2 / 17.4 / 17.7 | 22.8 / 20.6 / 21.2 | 10.1 / 8.7 / 8.1 | 9.5 / 8.0 / 10.5 | 2.8 / 2.4 / 2.6 |

(B3 is lowest) but the worst position error of the dynamic-anchor variants (15.2–16.3 m): the offsets are accumulated against the voted reference of Eq. (1) yet applied at the biased visible centroid, and the mismatch misplaces individual players. B4 applies the same EMA offsets at the voted centroid instead, making estimation and application self-consistent, and obtains the best position error. B3E and B3 are best read as diagnostic reference-mismatch variants — both estimate offsets against the voted centroid but apply them at the visible centroid — rather than as natural competing baselines. The clearest negative result is the fixed template (B5), far worse than any dynamic anchor (position error 20.7–22.7 m; share error 6.9–10.3 pp) — consistent with within-half role drift, though slow adaptation and accumulated subset bias may also contribute.

**Sensitivity.** Table 2 and Figure 1 report the sweep over viewport widths. Across $W = 36$–60 m B4 improves on B0 in every cell of all three matches, reducing the share error to 28–48% of B0 across games and widths, with the largest absolute share-error gains in the narrowest (hardest) viewports — the method is most valuable exactly when observation is poorest.

## 5.1 Occlusion-time stratification and protocol comparison

Table 3 contrasts the observation protocols, whose differences are likely favorable to the learned model — bidirectional inference over future observations, 9.6 s sequence segmentation, and 105

Table 3: Observation-protocol comparison with the Graph Imputer [10]. The regimes differ in the two dimensions that matter most — temporal information and training data — so numeric results are not directly comparable; the stratified analysis below partially narrows the occlusion-regime mismatch.

| | Graph Imputer | This work |
|---|---|---|
| Camera model | view-cone × pitch polygon, ball-led | ball-EMA pan window (width $W$) |
| Visible players | $12.8 \pm 3.7$ of 22 | 14.6–15.0 of 22 ($W = 44$ m) |
| Temporal structure | bidirectional (sees *future* obs.), sequences segmented at 9.6 s | online, causal, occlusions unbounded |
| Training data | 105 proprietary EPL matches | none (current match only) |
| Compute | GPU training | CPU, real time |

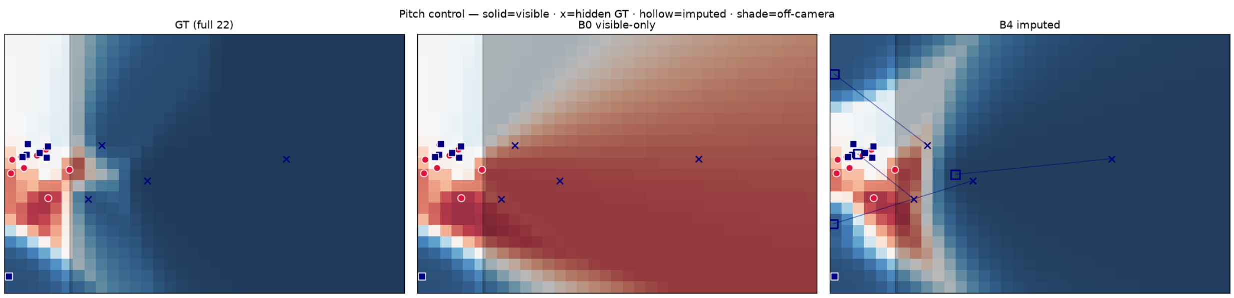


Figure 2: Benchmark visualisation (Metrica, $W = 44$ m). Left: ground-truth pitch control from all 22 players. Middle: control from visible players only (B0) — the hidden zone (outside the vertical viewport band) is systematically misassigned. Right: control with B4 imputation (ghost markers) — the hidden zone is largely restored without any training data.

matches of training data — so we align the comparison by stratifying B4's position error by occlusion duration:

| occlusion gap | ≤2 s | 2–9.6 s | >9.6 s |
|---|---|---|---|
| B4 median position error | 3.3–3.7 m | 7.2–8.9 m | 15.6–16.9 m |
| share of hidden samples (frame-weighted) | ≤9.6 s: 43–50% | | >9.6 s: 50–57% |

For occlusion gaps short enough to fit within a 9.6 s window — the regime closest to the Graph Imputer's protocol — a training-free online method reaches 3.3–8.9 m median error. We do not attempt a numeric comparison: masking, aggregation, and data all differ, and we do not claim superiority over learned bidirectional models within their own evaluation regime; the point is that a zero-training method is usable there at all. Roughly half or more of the hidden (frame, player) samples in our benchmark (50–57%, frame-weighted — long episodes dominate this count) lie *beyond* 9.6 s: long occlusions of far-side defenders during sustained attacking phases are precisely where spatial metrics need imputation most, and it lies outside the sequence protocol of the closest prior work. It is also where our absolute errors (15.6–16.9 m median) show the remaining headroom, which we expect learned long-horizon models to attack.

# 6 Application: Possession-Quality Verdicts on Real Broadcasts

To probe the effect on real footage, we integrate a B4-inspired ghosting layer into an open broadcast-video GSR pipeline: PnLCalib-based camera calibration [6] with a temporal bridge, BoT-SORT tracking [1], colour-based team assignment, and shot-change gating; the pipeline scores GS-HOTA 35.2 on the 11 publicly downloadable sequences of the SoccerNet-GSR test split (for context only: the official baseline reports 26.6 on the full split [12], so the two numbers are not strictly comparable; challenge-winning systems with GPU appearance re-identification score substantially higher, and our pipeline is CPU-only). Off-screen players are rendered as "ghosts" with role-anchored positions that fade over a 20 s lifetime; ghosts top the team up to at most eleven players (visible plus ghosts), and a ghost is retired when a newly appearing track (within 2 s of its birth) emerges within 3 m of the ghost's prediction. Role offsets are keyed to track identities, so a fragmented track restarts its offset estimate — the operational layer is B4-inspired rather than the benchmarked algorithm verbatim.

As a downstream consumer we use a possession-quality metric, the *Space-Creation Index* (SCI): during a flagged low-threat possession ("junk possession"), $\mathrm{SCI} = \Delta_{\mathrm{own}} + \Delta_{\mathrm{opp}}$, where $\Delta_{\mathrm{own}}$ is the change in the possessing team's pitch-control share of the attacking third between the first and last thirds of the window, and $\Delta_{\mathrm{opp}}$ is the collapse of the opponent's control share in its own advanced zone (both in percentage points, from the same pitch-control model as Section 3). Operational verdict classes: *space creation* (SCI $\geq +12$) — the possession creates space even without shots; *weak progression* ($+4 \leq$ SCI $< +12$); *dead possession* (SCI $< +4$). We evaluate the two junk-possession windows that our event-side index flags for a FIFA World Cup 2026 Round-of-32 match between the Netherlands and Morocco (1–1 after extra time; Morocco won 3–2 on penalties) — flagged from event data alone (possession sequences of negligible threat under an expected-threat grid), before any spatial analysis, and they are the only flagged windows for this match, so there is no post hoc window selection. Morocco is the possessing team in both. Window 1 spans game clock 11:58–12:32 (34 s), window 2 spans 72:48–73:45 (57 s); scorelines below are at each window's start. The attacking third and the opponent's advanced zone are the corresponding thirds of pitch length in each team's attack direction; shares are frame-averaged over the first and last temporal thirds of the window, and the SCI class thresholds (+12/+4) are prespecified operational bins, set in our earlier internal work on this index before these windows were analysed but not externally validated. Imputation moves the score by a decision-relevant magnitude:

- Window 1 (0–0, min. 12): SCI $+15.6 \to +32.8$. Under the imputation policy, the estimated off-screen defenders produce a substantially *stronger* space-creation signal than the visible-only estimate.
- Window 2 (0–1, min. 73): SCI $-10.9 \to +4.7$ — the verdict class changes from "dead junk" to "weak progression".

Absent full-pitch ground truth for broadcast footage we make no claim that the imputed scores are more accurate; the point is a sensitivity result. Visible-only spatial verdicts can move by 15.6–17.2 SCI points under imputation — more than spanning the 8-point intermediate verdict class — so spatial possession-quality reported from broadcast video without imputation is sensitive to camera framing and the chosen off-screen-player policy.

## 7 Limitations and Future Work

Identity fragmentation in real GSR output (track fragments; jersey-number OCR coverage ≈45% in our pipeline) makes roster-aware ghosting approximate on real video: ghosts anchor to roles, not to verified identities. Appearance re-identification is the natural next step. The pitch-control model is velocity-free here to isolate the imputation effect; velocity-aware control with imputed velocities is open — the ablation shows short-gap extrapolation is mildly informative for position, but its effect on velocity-aware control is untested. The benchmark uses three matches from a single provider; more matches and leagues would tighten the estimates, though the effect sizes are large and consistent across all three matches and every viewport width. The simulated viewport pans but does not zoom or tilt; zoom changes the number of visible players, so real-broadcast occlusion statistics (including the 50–57% long-occlusion share) may differ from the simulated ones. All hyperparameters ($\tau = 8\,\mathrm{s}$, EMA weight 0.1, the three-voter threshold, the 1.5 s blend constant) are round values fixed once during development on games 1–2 and not optimised by search; the held-out game 3, evaluated only after they were frozen, replicates the broad improvement over B0 and the effect magnitudes, though the share-error ranking tightens (B2/B3V edge B4 by 0.3 pp). Finally, learned imputers remain preferable when future context and training data are available — our contribution is the training-free floor, the harsher online/long-occlusion benchmark, and the downstream-metric framing.

## 8 Reproducibility

All benchmark experiments run on CPU with open data (Metrica Sports sample data [8]) and open code; the benchmark, the imputation ladder, the figures, and the exact commands are available at `https://github.com/nowayfootball/offscreen-impute`. The broadcast case study uses World Cup footage that we cannot redistribute. The benchmark code is publicly available, while the case-study pipeline and footage are not included in the public repository. No GPU or cloud expenditure was used for any experiment in this paper.

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